

OSMOTIC PRESSURE

What is diffusion?

The phenomenon of flowing of solute molecules from a solution of higher concentration to a solution of lower concentration (through a permeable membrane) until concentration of both solutions becomes equal is called diffusion.

The diffusion of solute into solvent is, in fact, a bilateral process. It consists of: (1) the solute molecules moving up into solvent; and (2) the solvent molecules moving down into solution. This intermingling of solute and solvent molecules goes on, so that ultimately a solution of uniform concentration results.

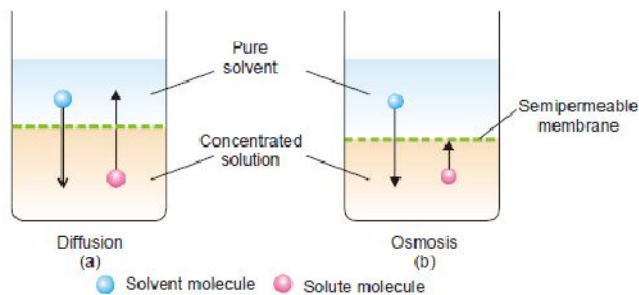


Fig.1:(a) Diffusion consists of solute molecules moving upward into the solvent and solvent molecules moving down into concentrated solution; (b) Osmosis consists of solvent molecules passing through the semipermeable membrane, while large solute particles are held up.

What is osmosis?

The flow of the solvent through a semipermeable membrane from pure solvent to solution, or from a dilute solution to concentrated solution, is termed Osmosis.

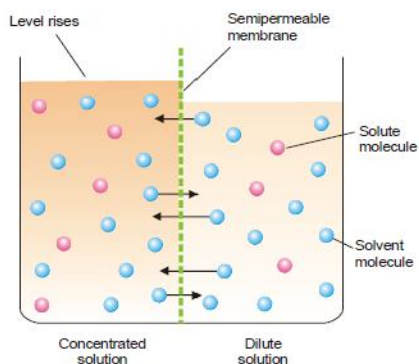


Illustration of the process of osmosis. The solvent molecules can pass through the membrane from either side of the semipermeable membrane. There is a net flow of solvent from the dilute solution to the concentrated solution. It shows up in the rise of level in the concentrated solution compartment.

Difference between diffusion and osmosis

Osmosis

1. Migration of solvent molecules from a solution of lower concentration to a solution of higher concentration until equilibrium is attained.
2. Solvent molecules pass through a semi permeable membrane.
3. Osmosis is a slow process which may take even a few days.

Diffusion

1. Migration of solute molecules from a solution of higher concentration to that of lower concentration until equilibrium is obtained.
2. Solute molecules pass through a permeable membrane.
3. Diffusion is a rapid process which may be completed in a few minutes.

Osmotic pressure:

The hydrostatic pressure built up on the solution which just stops the osmosis of pure solvent into the solution through a semipermeable membrane, is called Osmotic Pressure.

Osmotic pressure may be defined as the external pressure applied to the solution in order to stop the osmosis of solvent into solution separated by a semipermeable membrane.

Isotonic solution

When two solutions are separated by a semipermeable membrane and there is no flow of water across the membrane, the solutions are said to be Isotonic. The isotonic solutions have equimolar concentrations.

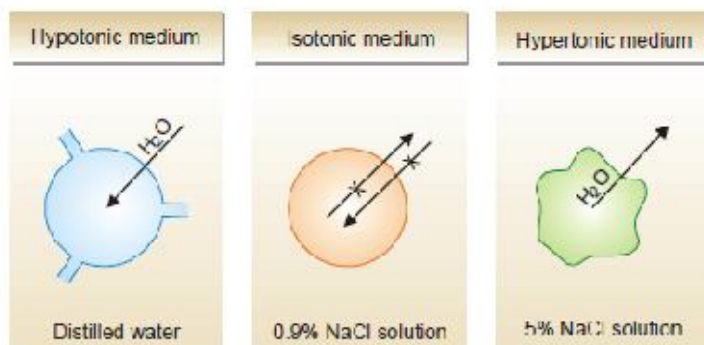


Fig.4 Osmotic effects on Red Blood Corpuscle

Hypotonic solution

Of the two solutions separated by a semipermeable membrane if one is of lower osmotic pressure, it is said to be hypotonic relative to the second solution.

Hypertonic solution

If it has a higher osmotic pressure than the second solution, it is said to be **hypertonic** relative to the second solution.

When red blood cells are placed in distilled water (*hypotonic medium*), water flows into the cells and they swell or burst. If, on the other hand, they are placed in 5% NaCl solution (*hypertonic medium*), water comes out of the cells and they shrink. A 0.16 M sodium chloride solution (0.95%) is isotonic with blood cells and they neither swell nor shrink as no osmosis takes place.

Semi-permeable membrane

The membrane through which solvent molecules can pass from solvent to solution side or from solution of lower concentration to solution of higher concentration is called semi-permeable membrane.

Examples: Fish bladder, Pig's bladder, Lining of egg shell, many inorganic precipitates like-copper ferrocyanide.

Permeable membrane: A membrane through which solute molecules can easily pass.

Methods for determination of osmotic pressure

1. Pfeffer's method
2. Barkley and Hartley's method
3. By an osmometer

Berkeley and Hartley's Method

Berkeley and Hartley (1904-1909) employed the technique of applying external pressure on the solution just enough to prevent osmosis. The osmometer used by them is illustrated in Fig.. A porcelain tube with copper ferrocyanide membrane deposited in its walls is enclosed in a metallic jacket. The tube is fitted with a reservoir of pure solvent (water) at one end and a capillary tube at the other. Mechanical pressure can be applied on the solution with a piston connected to a pressure gauge.

Procedure. The inner porcelain tube is filled with pure solvent and the jacket with the solution whose osmotic pressure is to be determined. The level of the solvent meniscus in the capillary tube will tend to move down as solvent flows into the solution across the membrane. Pressure is then applied through the piston so that the meniscus becomes stationary. It indicates that osmosis has been stopped and now the pressure recorded by the pressure gauge gives the osmotic pressure of the solution.

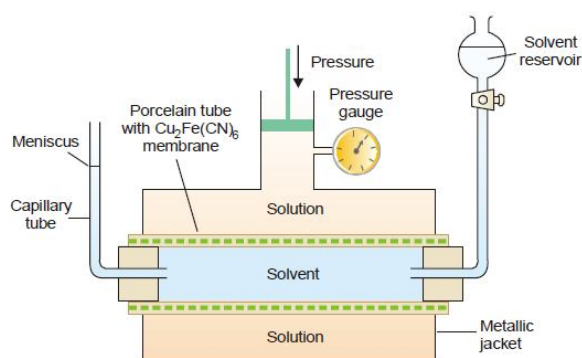


Fig. Berkeley and Hartley's Method

Reverse osmosis:

The osmosis taking place from solution to pure water by application of pressure greater than osmotic pressure, on the solution, is termed Reverse Osmosis.

Desalination of Sea Water by Hollow-fiber Reverse Osmosis

Reverse osmosis is used for the desalination of sea water for getting fresh drinking water. This is done with the help of hollow fibres (nylon or cellulose acetate) whose wall acts as semipermeable membrane.

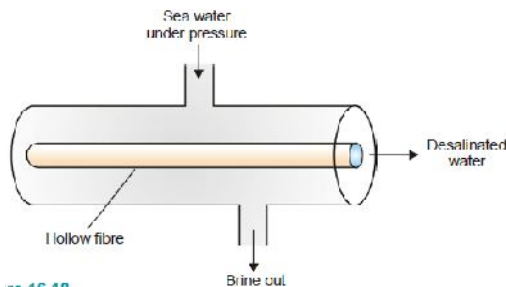


Fig. 5. Desalination of sea water by reverse osmosis in a hollow fibre unit.

A hollow-fibre reverse osmosis unit is shown in Fig. 5. Water will introduced under pressure around the hollow fibers. The fresh water is obtained from the inside of the fibre.

Laws of osmotic pressure

(1) Boyle-van't Hoff Law for Solutions

The osmotic pressure of a solution at a given temperature is directly proportional to its concentration. If Π is the osmotic pressure and C its concentration, we can write $\Pi \propto C$, if temperature is constant. If the concentration of the solute is expressed in moles per litre and V is the volume of the solution that contains 1 mol of solute, $C = \frac{1}{V}$, thus $C \propto \frac{1}{V}$ at constant temperature.

This relationship is similar to the *Boyle's law* for gases and is known as the *Boyle-van't Hoff law for solutions*.

(2) Charles-van't Hoff Law for Solutions

The osmotic pressure of a solution of a given concentration is directly proportional to the absolute temperature.

If T is the absolute temperature, we can write $\Pi \propto T$, if temperature is constant. This relationship is similar to the *Charles' Law for gases* and is known as *Charles-van't Hoff law for solutions*.

Van't Hoff Equation for Solutions

As the osmotic pressure (Π) of a dilute solution is inversely proportional to the volume (V) containing 1 mole of the solute and is directly proportional to the absolute temperature (T).

This is, $\Pi \propto C \dots (1)$

$\Pi \propto T \dots (2)$

Combining (1) and (2) van't Hoff gave the general relationship

$\Pi V = R' T \dots (3)$

where R' is a constant. He showed that this equation was parallel to the general *Gas Equation* ($PV = RT$), as the value of R' calculated from the experimental values of Π , V , and T came out to be almost the same as of the *Gas constant*, R .

It is noteworthy that the *van't Hoff Equation* (3) was derived for 1 mole of solute dissolved in V litres. If n moles of solute are dissolved in V litres of solution, this equation assumes the form

$\Pi V = n R T$

Third law: Equimolecular quantities of different solutes dissolved in same volume of solvent so as to give same volume of solution exert equal osmotic pressure under identical conditions of temperature.

Problem

A solution containing 10.2g glycerine solution per litre is found to be isotonic with a 2% solution of glucose. Calculate molecular weight of glycerine when molecular weight of glucose is 180.

For glycerine solution $P_1 V_1 = nRT = \frac{g}{M} RT$ ($n = \frac{g}{M}$)

Or, $P_1 = \frac{gRT}{M_1 V_1} = \frac{10.2 RT}{M_1}$

Again, $P_2 V_2 = \frac{g}{M_2} RT$ for glucose

So, $P_2 = \frac{gRT}{M_2 V_2} = \frac{20 RT}{180}$

As the two solutions are isotonic that means, their osmotic pressures are same So, $P_1 = P_2$

$\frac{10.2 RT}{M_1} = \frac{20 RT}{180}$

Or, $M_1 = 91.8 \sim 92$

So, the molecular weight of glycerine is 92.

VAN'T HOFF THEORY OF DILUTE SOLUTIONS

A substance in solution behaves exactly like a gas and the osmotic pressure of a dilute solution is equal to the pressure which the solute would exert if it were a gas at the same temperature occupying the same volume as the solution.

From van't Hoff theory it follows that just as 1 mole of a gas occupying 22.4 litres at 0°C exerts 1 atmosphere pressure, so, 1 mole of any solute dissolved in 22.4 litres would exert 1 atmosphere osmotic pressure.

van't Hoff theory holds only for dilute solutions and if there is no dissociation or association of the solute molecules.

Haemolysis

If a red blood corpuscle is immersed in a hypotonic media that is in such a solution whose concentration is lower than the inner concentration of the cell, then water will enter in to inside of the cell, as a result it will swell and finally cause it to burst-this phenomenon is known as haemolysis.

Plasmolysis

If a red blood corpuscle is immersed in a hypertonic media that is in such a solution whose concentration is higher than the inner concentration of the cell, then water will pass out through the cell and consequently there will be partial collapse of the cell membrane causing a change in the shape of cytoplasm. The cell is then said to be plasmolyzed and the phenomenon is known as Plasmolysis.